# Neon Electroporation Transfection

FV01 | ver 1.1 | Revised 2026-10-03

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## Purpose

Electroporate CCF-STTG1 cells using the Neon system at 1350 V, 20 ms and 2 pulses, with 200,000 cells per 10 µL reaction. The cell input and pulse program are the team-specified working conditions. The plasmid preparation and recovery guidance below use the manufacturer's general recommendations.

## Materials & Equipment

| Item | Supplier / model | Use / specification |
|---|---|---|
| CCF-STTG1 cells | Laboratory culture | 200,000 cells per loaded 10 µL reaction |
| Electroporation system | Invitrogen Neon | Original Neon device, pipette and pipette station |
| 10 µL tips and Neon tubes | Invitrogen Neon 10 µL kit | Compatible kit options: MPK1025 or MPK1096 [1] |
| Resuspension Buffer R | Supplied Neon reagent | Cell suspension inside the 10 µL tip [2] |
| Electrolytic Buffer E | Supplied Neon reagent | 3 mL in the external Neon tube; separate from the cell mixture [1,2] |
| Plasmid DNA | Purified experimental / control construct | Manufacturer reference: 0.5-3 µg per reaction for the 12-well setup; stock 1-5 µg/µL; DNA solution ≤1 µL per 10 µL reaction [1] |
| PBS; 0.25% Trypsin-EDTA | Gibco / laboratory stocks | Ca2+/Mg2+-free PBS; adherent-cell preparation as in C01 |
| Recovery medium; 12-well plate | Gibco RPMI 1640 + Cellmax FBS | C01 formulation without penicillin-streptomycin: 500 mL RPMI 1640 + 50 mL FBS; reference recovery volume 1 mL/well [1] |
| Counter, centrifuge, microscope; sterile tubes and pipettes | Laboratory equipment | Cell counting, harvest and recovery assessment |
| Cell culture cabinet; humidified incubator | Laboratory equipment | Aseptic work; 37°C, 5% CO2 as in C01 |

## Before You Start

Prepare healthy CCF-STTG1 cultures using C01. Prewarm antibiotic-free recovery medium and prepare one labeled 12-well well containing 1 mL per electroporation. This is the manufacturer-based reference format for 200,000 adherent cells, not a claim about a previously performed run. [1]

Program the Neon device to 1350 V, 20 ms, 2 pulses. Use Buffer R for this workflow. The pulse settings were supplied by the team; they are not presented as a manufacturer-validated CCF-STTG1 optimum.

For plasmid delivery, select a dose within the 0.5-3 µg reference range according to the construct and endpoint. Use measured stock concentration to calculate volume; keep the combined volume of all DNA stocks ≤1 µL. For co-transfection, apply the selected construct ratio within the total dose. [1]

Include untreated cells, a pulse-only control and a suitable delivery-positive control when establishing the method. Add an empty-vector or non-targeting control appropriate to the functional comparison. Select the readout time for the expressed product and downstream assay.

## Procedure

### A. Define the 10 µL reaction

| Parameter | Per electroporation |
|---|---|
| Loaded cell number | 200,000 cells |
| Loaded mixture / tip size | 10 µL / 10 µL Neon tip |
| Final cell density | 20,000 cells/µL = 2 × 10^7 cells/mL |
| Plasmid DNA | 0.5-3 µg reference range; VDNA = dose (µg) / stock concentration (µg/µL); VDNA ≤1 µL [1] |
| Cells in Buffer R + DNA | Vcells + VDNA = 10 µL |
| Required cell-stock density | 200,000 / Vcells cells/µL |
| Pulse program | 1350 V / 20 ms / 2 pulses |
| Recovery reference | One 12-well well containing 1 mL antibiotic-free medium [1] |

1. Calculate VDNA from the selected dose and measured stock concentration. Prepare the cell suspension in Buffer R at 200,000 / (10 - VDNA) cells/µL. Scale both cells and DNA proportionally for any pipetting excess; each loaded 10 µL must contain 200,000 cells.

Calculation example: 1 µg DNA at 1 µg/µL occupies 1 µL. Combine it with 9 µL of cells in Buffer R at approximately 22,222 cells/µL. This gives 200,000 cells in 10 µL. The example is not a measured optimum.

### B. Prepare cells and the station

2. Detach CCF-STTG1 cells using C01 and count them. Pellet at 100-400 × g for 5 min at room temperature; wash once with Ca2+/Mg2+-free PBS using the same centrifugation conditions. [1]
3. Remove PBS and gently resuspend the pellet in Buffer R at the calculated cell-stock concentration. Keep the suspension uniform for dispensing. [2]
4. Add 3 mL Buffer E to a Neon tube and seat it in the station. Verify 1350 V, 20 ms and 2 pulses on the display. [1]

### C. Load, pulse and recover

5. Mix the calculated DNA and cell-suspension volumes. Attach a 10 µL Neon tip securely and slowly aspirate 10 µL without bubbles. [1]
6. Seat the pipette vertically in the station and run the program. Once complete, immediately transfer the cells into the prepared well containing 1 mL prewarmed antibiotic-free medium. [1]
7. Distribute cells gently and incubate at 37°C with 5% CO2. Record construct identity, total DNA dose, actual pulse program, cell input, recovery well and any arcing.
8. At the predefined endpoint, assess delivery and viability separately; compare the experimental and control groups before interpreting downstream functional changes.

## p.s.

Buffer R belongs in the cell mixture; Buffer E belongs in the external Neon tube. Do not replace one with the other. [2]

Process small batches; avoid holding cells in resuspension buffer beyond 15-30 min. [1]

Discard a bubble-containing load and reload fresh mixture before pulsing. [1]

Use a new tip and tube for each different plasmid sample; maximum reuse is two runs per tip and ten per tube. [1]

Adding material to an already prepared 10 µL cell aliquot dilutes the cell input in the portion actually loaded. Use the final-mixture calculation above.

## Expected Outcome

Assess plasmid delivery and viable-cell recovery separately, then evaluate the intended biological endpoint at the selected time. Compare with the relevant controls before interpreting functional changes; this protocol does not prescribe a universal efficiency or viability threshold.

## Troubleshooting

| Observed problem | Check first | Action |
|---|---|---|
| Arcing / sparks | Bubbles, buffer choice and loading [1] | Flag the affected sample. Correct loading and buffer compatibility before repeating with fresh cells. |
| Low cell recovery | Starting viability; actual pulse program; handling times; pulse-only control | Identify whether loss also occurs without material. Reassess cell preparation and recovery conditions before changing the material dose. |
| Low delivery with acceptable recovery | Material identity, concentration, dose and positive-control result | Verify material quality and assay timing, then evaluate a cell-specific pulse optimization plan. |
| Unequal cell input across samples | Count, mixture volume and settling during preparation | Recalculate final density and gently remix before each load; scale all components together. |

## Reference

[1] Thermo Fisher Scientific / Invitrogen. Neon Electroporation System User Guide, MAN0001557, Rev. D, 27 June 2024. Kit contents, DNA preparation and electroporation, pp. 8, 23-32. Accessed 2026-10-03. Original Neon system. Open source

[2] Thermo Fisher Scientific / Invitrogen. Neon Transfection System 10 µL Kit, MPK1025, product FAQs: resuspension versus electrolytic buffers. Accessed 2026-10-03. Open source